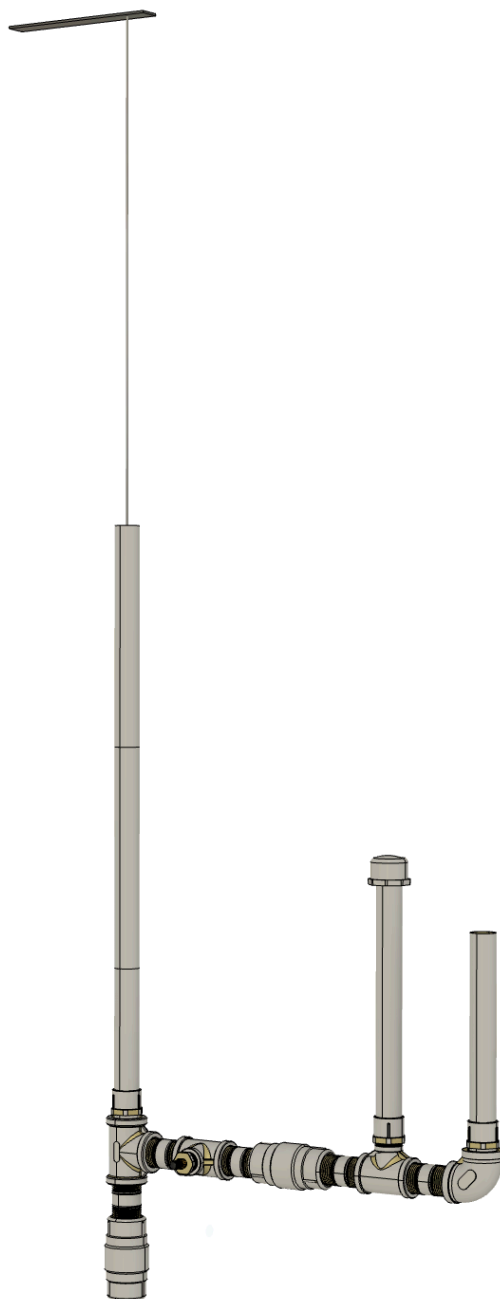


# Manual for the AguaClara Vertical Ram Pump

**By: Ram Pump Subteam SP2026**

**Katie Kim, Isaac Jiang, Jesús Paz, Alma Meili Ren**



*Figure 1.*

**Purpose of the System:**

The purpose of the AguaClara Vertical Ram Pump (ACVRP) is to transport a portion of treated water from the end of the AguaClara treatment process from the ground floor to the top of the plant, where the water will be used to dilute the chemical stock solution. This eliminates the need for plant operators to manually transport water to the chemical storage tanks. The pump is designed to take advantage of the physical properties of fluids and the conservation of energy, ultimately resulting in a size- and cost-efficient method of sending water to a higher elevation without the use of electricity. Once in operation, the ACVRP is estimated to pump water to a height of about 7 meters at more than 28mL/s.

**AguaClara Vertical Ram Pump Materials:**

- Check Valve (2): 1" Campbell Check Valve, one valve modified with an eyebolt in lab
- Eyebolt (1): M5 x 25mm Stainless Steel Eyebolt
- Main Tee (1): 1" 304 Stainless Steel Tee, Female Threaded
- Pipe Nipple (5): 1" 304 Stainless Steel Nipple (Standard Wall)
- Inline Reducer Tee (1): 1" & 1/2" Stainless Steel Reducer Tee
- Reducer Bushing (1): 1/2" to 1/8" Stainless Steel Bushing
- Snifter Valve (1): 1/8" MIP Brass Snifter, No-Spring Model
- Pressure Plug (1): 1/8" Stainless Steel Plug (for sealing the snifter valve port during system testing or control runs)
- MNPT Adapter (3): 1 in. x 1 in. CPVC CTS Slip Stainless MNPT
- Air Chamber Pipe (1): 1" Clear PVC Pipe, 42" Length
- Air Chamber Cap (1): 1" PVC Cap

- Flat Spring (1): Stainless Steel Metal Spring, fabricated in lab
- C-Clamp (1): Sliding T-Handle, 0" - 1" Opening
- Drive Pipe (1): 1" Threaded PVC Pipe
- Strut Channel (1): Steel Strut Channel, 3" Length,  $\frac{3}{8}$ " Wide Slotted Holes
- Wedge Anchor (2): Vibration-Resistant Wedge Anchor for Concrete,  $\frac{3}{8}$ " Diameter, 3.5" Length
- Teflon Tape (3)

### **Installation Process of the System:**

1. Select the access hole in the filtration tank:
  - a. Identify a hole that has a diameter of at least 1" in the bottom of the filtration tank that provides access to the basement.
  - b. Confirm that there is at least 2 meters of vertical clearance below this hole.
  - c. If the identified hole is larger than 1", use PVC reducer(s) to reduce it to 1".
  - d. Ensure that this hole is **not** required for any other plant operations as it will be inaccessible once the ram pump is installed.
2. Install the vertical PVC pipe:
  - a. From the top of the tank, insert a 1" PVC pipe into the selected hole, ensuring the top of the pipe remains above the water level to prevent water from spilling into the basement during installation.
  - b. From the basement side, pipe cement a ~1.8 m length of 1" PVC pipe into the underside of the hole.
3. Thread the metal wire:

- a. Tape one end of the steel wire to a long pipe or rod.
  - b. Feed the rod through the 1" hole from the tank floor to the basement to thread the wire through the installed PVC pipe from Step 2b.
4. Prepare the ram pump connection:
- a. Assemble the ram pump components (tees, fittings, reducers, snifter valve, elbow, and MNPT adapters) as shown in the steps visualized below with Teflon tape.
  - i. Do **not** install the Vertical Check Valve or the Air Chamber.
  - ii. Ensure that the chamber valve is oriented so that the internal spring pushes the door closed towards the side of the ram pump with the snifter valve.
  - iii. Ensure that the snifter valve is facing horizontally, perpendicular to the drive pipe and air chamber.

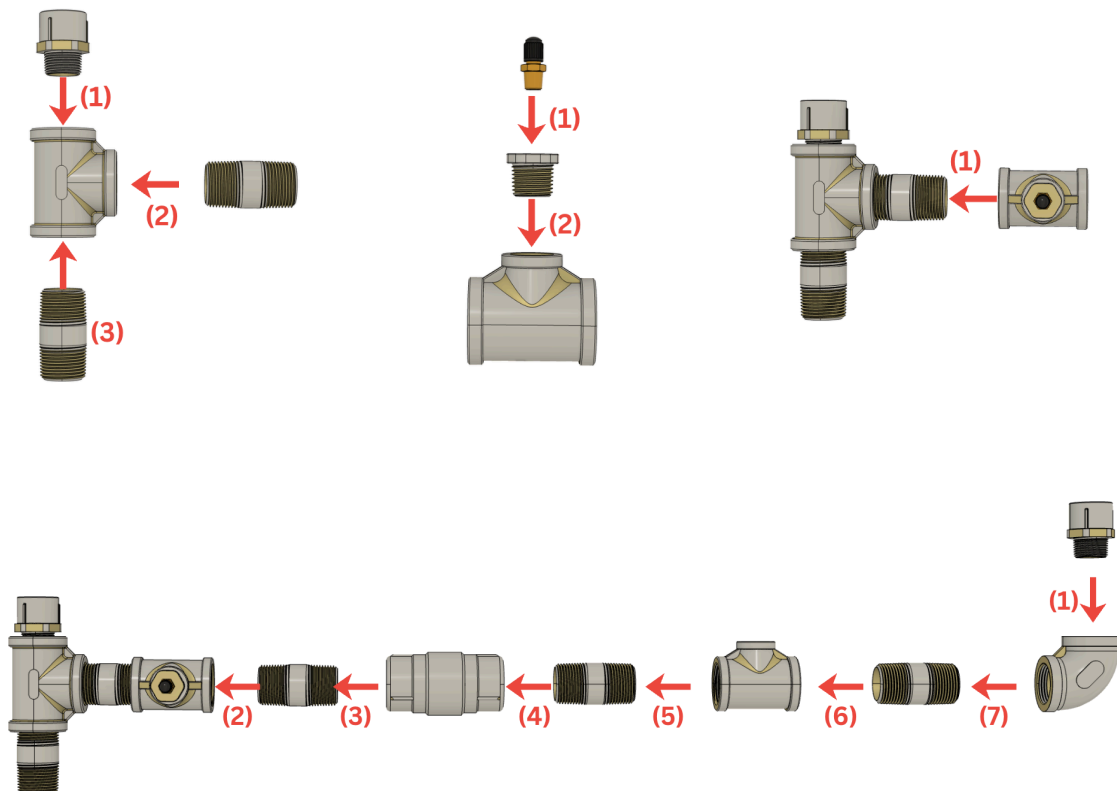


Figure 3.

5. Modify the Check Valve:

- a. Remove the internal spring by pulling it out with needle-nosed pliers.
- b. Vertically drill ~10 mm into the check valve door with a 4.2 mm drill bit.
- c. Tap this hole with a  $M5 \times 0.8$  tap.
- d. Install the Eyebolt into this hole with weld glue to ensure this connection doesn't come loose.
- e. Tighten the nut downward onto the check valve door.

6. Install the ram pump assembly:

- a. Pipe cement the MNPT adapter above the tee on the opposite side of the ram pump as the elbow to the bottom of the drive pipe, which is located in the basement.
  - i. Ensure that the ram pump is oriented so there are nearby pipes or ledges it can be anchored/tied to for support.

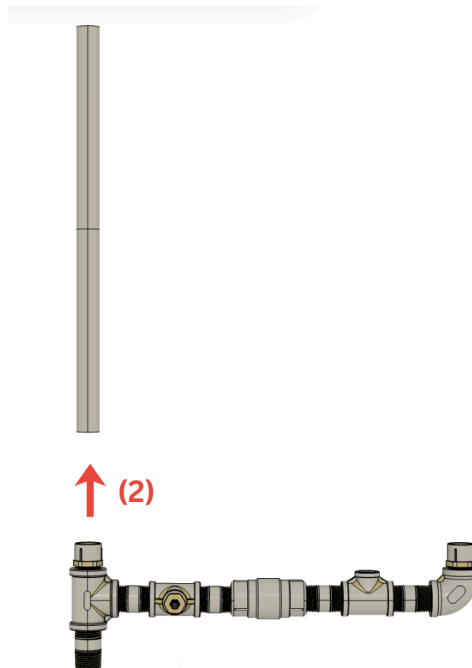
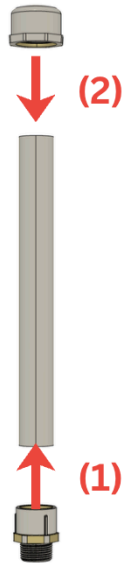


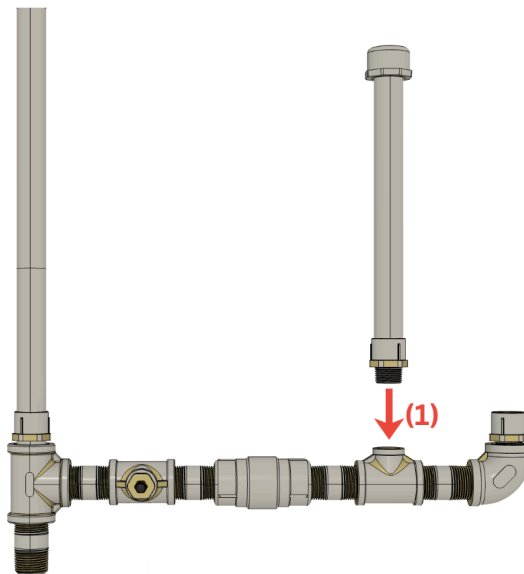
Figure 4.

- b. Pipe cement a PVC cap onto one end of the 1" PVC pipe cut to be the air chamber. Then, pipe cement a MNPT adapter to the other of this PVC.



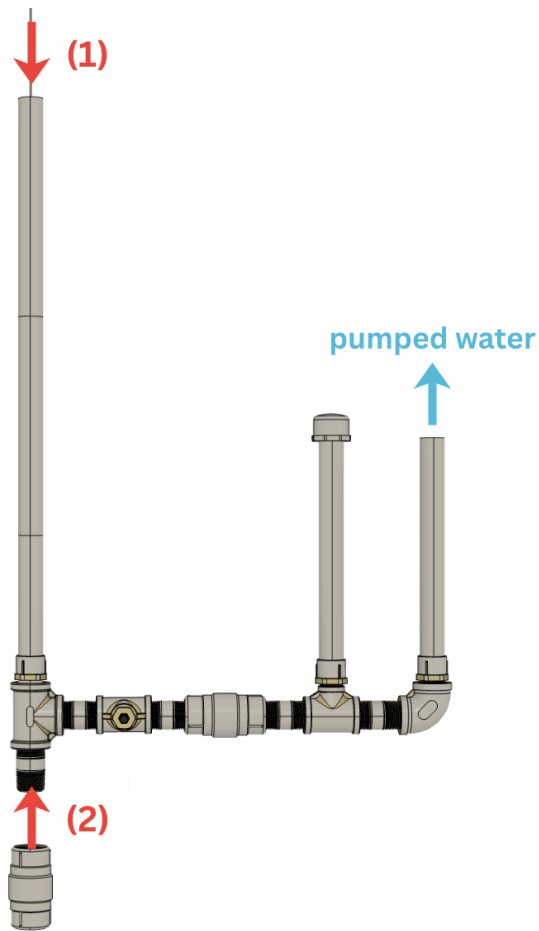
*Figure 5.*

- c. Install the constructed air chamber into the tee next to the elbow in the ram pump, ensuring that it is parallel to the drive pipe.



*Figure 6.*

- d. Thread the steel wire through the installed ram pump section and crimp it to the eyebolt on the modified Campbell Check Valve. Then, install the vertical Campbell Check Valve while a second person pulls the wire taut from the ground floor.



*Figure 7.*

7. Install the flat spring:
  - a. On the ground floor, mount the flat spring to the strut channel using a C-Clamp.
  - b. Temporarily secure the strut channel onto a tank wall with bricks or another heavy object during the spring calibration process.
  - c. Attach the hook to the end of the flat spring using nuts above and below the spring.



Strut installed on ground floor with wire leading down to the basement.

*Figure 8.*

8. Set the wire length:
  - a. Find the wire length at which when the flat spring is fully in the open state, the check valve door is completely open (at which point the wire cannot be pulled up any more).
  - b. Add some more length to the wire that can be used to make the loop that will be hooked onto the flat spring.

- c. Cut the wire at this length and crimp a small loop, ensuring that the size of this loop, when hung onto the flat spring hook, will again ensure that the waste valve is completely open and can't be pulled up any further.
      - i. You can also adjust the hook height by loosening the nuts until this condition is met and retightening, in the case that when the wire loop is hooked onto the flat spring the wire is a little too short or long.
    - d. Wrap tape around the hook below the lower nut to prevent loosening during operation.
- 9. Calibrate the system:
  - a. Adjust the spring until the ram pump produces continuous, consistent oscillations.
    - i. For a 2 cm wide flat spring a good working spring length to start calibration at is ~16cm.
  - b. Confirm that holding the spring down stops the system in the closed state and pulling the spring back up allows the system to restart normally.
    - i. If this behavior doesn't occur, increase the length of the flat spring so that holding down the spring will stop the operation of the ram pump.
  - c. Refer to the Calibration of the System section for more details.
- 10. Finalize alignment and mounting:
  - a. Once the ideal spring length is set, remove the bricks.
  - b. Slide the strut channel forward or backward so the wire runs straight down the PVC pipe without contacting the sides.
  - c. Run the system one final time to confirm smooth operation.

- d. Once you have smooth operation, detach the cable from the flat spring to carefully slide out the PVC pipe previously used to plug the opening. Since the new check valve will now handle the water seal, this pipe is no longer necessary. Once removed, reconnect and readjust the cable to the flat spring.
- e. Permanently mount the strut channel to the concrete wall of the filtration tank by drilling two holes and installing Wedge Anchors.



*Figure 9.*

### **Calibration of the System:**

There may be times when the system fails to run. However, the configuration of the pump allows for simple solutions to system failure, resulting in a self-sufficient system that requires minimal attention from plant operators in order to run continuously, even if failure occurs.

System failure will most likely be a result of the system's calibration—more specifically, the length of the spring and how it affects the spring's resilience. The resilience of a spring refers

to its ability to return to its original state. The spring's resilience can either be too high or too low. There are two possible failing positions of the spring: the "open" state and "closed" state.

If the spring is in the "open state," it will appear horizontally straight. This means that the spring resilience is too high compared to the force of the water on the waste valve, which will not be able to close. In order to fix this, we can make the spring less resilient by increasing the length of spring that is hanging over the ledge.

If the spring is in the "closed state," it will curve downward. This means that the spring resilience is too low to overcome the force of water on the waste valve, which will not be able to open. In order to fix this, we can make the spring more stiff by decreasing the length of spring that is hanging over the ledge.

#### **Procedure for System Failure: <sup>1</sup>**

1. Identify the failure state: determine whether the failure state is "open" or "closed."
  - a. "Open" Failure State: The waste valve remains open and water flows out continuously. This occurs when the spring is too stiff or has too much leverage for the water's velocity to overcome.
  - b. "Closed" Failure State: The waste valve remains shut and the pump stops cycling. This occurs when the spring is too weak to pull the valve back up against the static pressure.
2. Adjust the Flat Spring position: <sup>2</sup>
  - a. For an "open" failure state:
    - i. Loosen the spring clamp.

- ii. Slide the spring outward (away from the pump body). This increases the stiffness, allowing the water's force to close the valve more easily.
  - iii. Secure the spring clamp.
- b. For a "closed" failure state:
  - i. Loosen the spring clamp.
  - ii. Slide the spring inward (toward the pump body). This increases the stiffness and increases the spring's effective force to pull the valve open.
  - iii. Secure the spring clamp.

**Notes:**

<sup>1</sup> This calibration process may take multiple iterations of sliding the spring inward or outward before the system maintains a consistent, smooth cycle.

<sup>2</sup> This adjustment may require more sliding than initially expected. Begin by adjusting the flat spring position by ~1 cm. If the desired behavior is not achieved, increase the adjustment to approximately ~3 cm.

## Appendix

### Complete System:

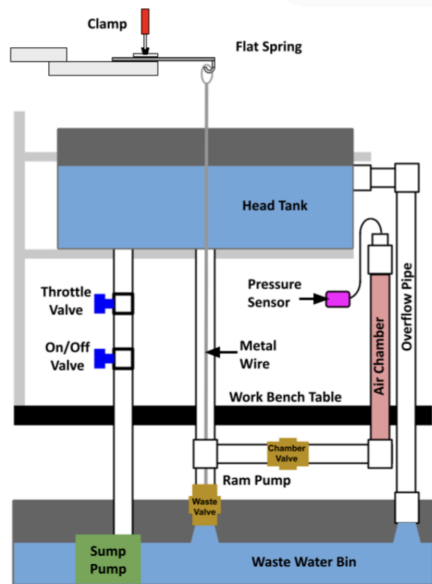


Figure 11. Schematic Diagram of Lab Setup

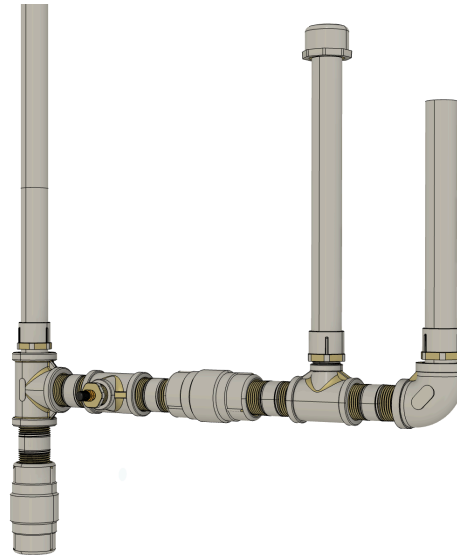


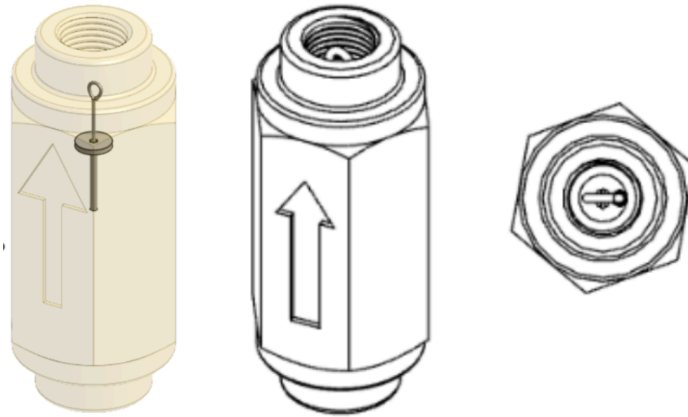
Figure 12: 3D Model of Field Installation Design

### Strut Intersection:



Figure 13. 3D Model of Flat Spring Mounting System

**Modified Campbell Check Valve:**



*Figure 15. 3D Model of Waste Campbell  
Check Valve*